Trusting Your Evidence: Hallucinate Less with Context-aware Decoding

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Background

- **Language models (LMs)**
  - Language models (LMs) are effective in generating fluent continuations of a prompt or document prefix
  - During generation, they rely on two sources of knowledge:
  - (1) Prior Knowledge
    - It is learned during pretraining and stored implicitly within the model parameters
  - (2) Context Knowledge
    - It is passed as inputs in the prefix context (Chan et al., 2022)
  - However, it remains an open question how a pretrained LM, particularly a vanilla LM without task-specific finetuning, balances these two knowledge sources during generation

Stephanie Chan et al. Data distributional properties drive emergent in-context learning in transformers. NeurIPS. 2022.

Background

- **Insufficient attention to context is problematic**
  - LMs can fail to pay enough attention to new information introduced in the context knowledge
  - Intrinsic hallucination: Manipulate the information present in the input document

Focus ➡ **Extrinsic hallucination: Add information not directly inferable from the input document**

Hallucinations (Maynez et al., 2020) in extreme document summarization (XSUM, Narayan et al., 2018a)

GOLD	Zac Goldsmith will contest the 2016 London mayoral election for the Conservatives, it has been announced.	
PTGEN	UKIP leader Nigel Goldsmith has been elected as the new mayor of London to elect a new Conservative MP.	[45.7, 6.1, 28.6]
TCONVS2S	Former London mayoral candidate Zac Goldsmith has been chosen to stand in the London mayoral election.	[50.0, 26.7, 37.5]
TRANS2S	Former London mayor Sadiq Khan has been chosen as the candidate to be the next mayor of London.	[35.3, 12.5, 23.5]
GPT-TUNED	Conservative MP Zac Goldwin's bid to become Labour's candidate in the 2016 London mayoral election.	[42.4, 25.8, 36.4]
BERTS2S	Zac Goldsmith has been chosen to contest the London mayoral election.	[66.7, 40.0, 51.9]

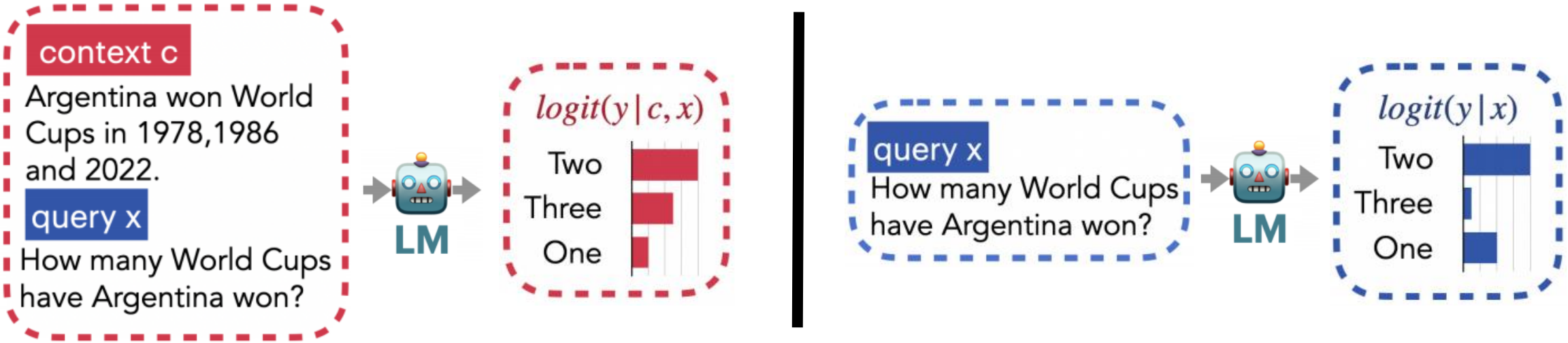
Add new info

Joshua Maynez et al. On faithfulness and factuality in abstractive summarization. ACL. 2020.
Shashi Narayan et al. Don't give me the details, just the summary! Topic-aware convolutional neural networks for extreme summarization. EMNLP. 2018.

Introduction

- **Insufficient attention to context is problematic**
  - Case: the context knowledge contradicts with the prior knowledge

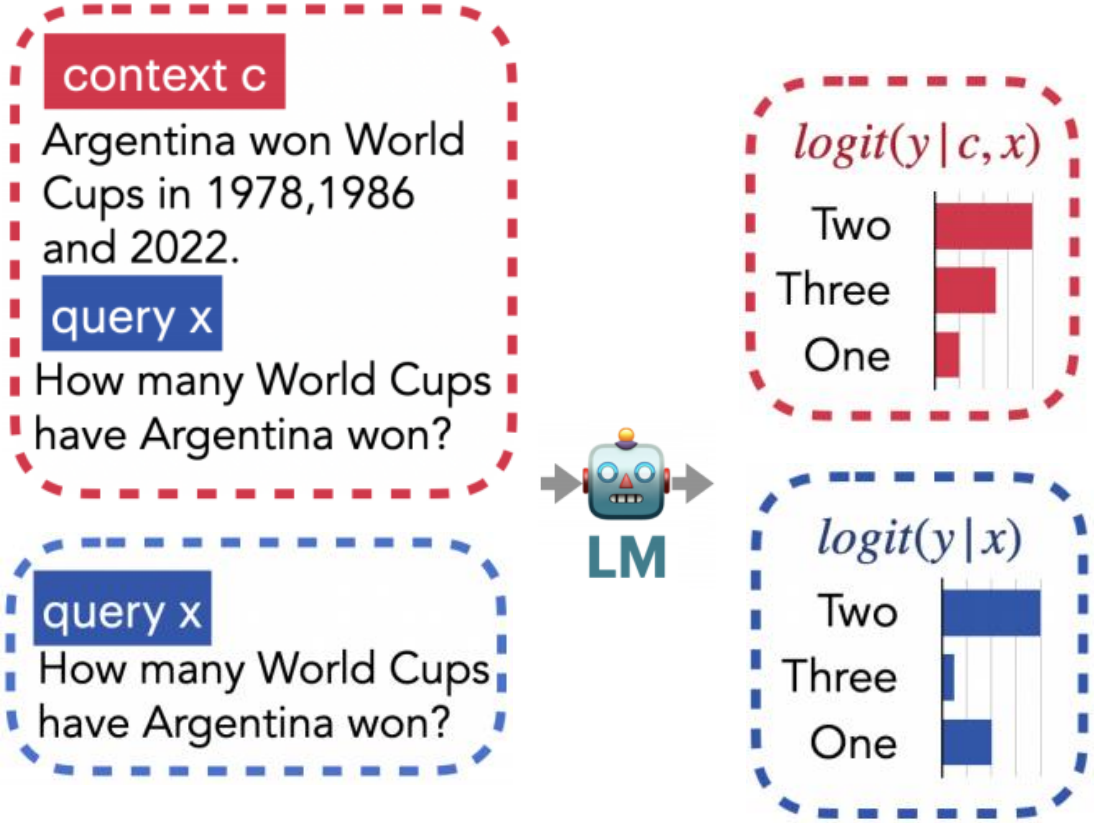
LLaMA (Touvron et al., 2023)



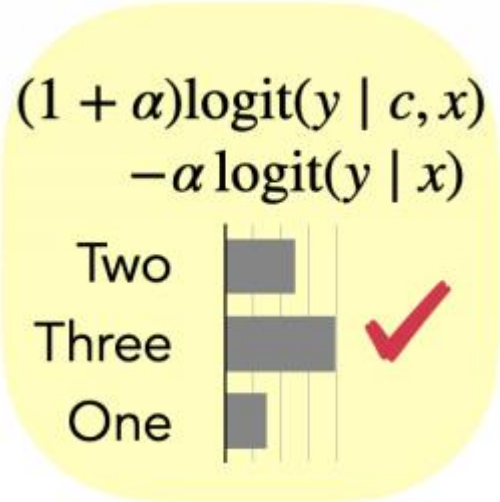
Hugo Touvron et al. LLaMA: Open and efficient foundation language models. ArXiv preprint, abs/2302.13971. 2023.

Introduction

- **Context-aware decoding (CAD)**
  - Effectively downweights the prior knowledge when more relevant contextual information is provided
  - It is used with off-the-shelf pretrained language models without any additional training



A new form of contrastive decoding (Li et al., 2023)



Xiang Lisa Li et al. Contrastive decoding: Open-ended text generation as optimization. ACL. 2023.

Method

- **Background**
  - A context $\mathbf{c}$ contains some external knowledge unfamiliar or in conflict to the model's prior knowledge
  - Ask our model $\boldsymbol{\theta}$ to generate a response $\mathbf{y}$ given the the query $\mathbf{x}$ and context $\mathbf{c}$

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \propto \exp \operatorname{logit}_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})$$

  - If the context $\mathbf{c}$ contains knowledge that is out-of-distribution with respect to $\boldsymbol{\theta}$
  - Hypothesize that the model can struggle to effectively attend to $\mathbf{c}$ and overly rely on the prior knowledge encoded in $\boldsymbol{\theta}$

Method

- **Mitigate such issues**
  - Model the prior knowledge as $p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})$
  - Adjust the model's original output probability distribution using the pointwise mutual information (PMI) between the context $\mathbf{c}$ and the generation $\boldsymbol{y_t}$, conditioned on $\mathbf{x}$, $\boldsymbol{y_{<t}}$

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

Method

- **Mitigate such issues**
  - Pointwise Mutual Information (PMI)
  - Product-of-Experts (PoE) (Hinton et al., 2002)
  - Classifier-free diffusion models (Ho and Salimans, 2022)
  - Contrastive decoding (Li et al., 2023)

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \quad \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

Hinton. Training Products of Experts by Minimizing Contrastive Divergence. Neural Comput. 2002.
Jonathan Ho and Tim Salimans Classifier-free diffusion guidance. NeurIPS Workshop. 2021.
Xiang Lisa Li et al. Contrastive decoding: Open-ended text generation as optimization. ACL. 2023.

Method

- **Pointwise Mutual Information (PMI)**
  - A measure of association
  - It compares the probability of two events occurring together to what this probability would be if the events were independent

$$PMI(A, B) = \log \frac{p(A, B)}{p(A)p(B)}$$

  - It can take positive or negative values, but is zero if A and B are independent.

Method

- **Pointwise Mutual Information (PMI)**

$$PMI(A, B) = \log \frac{p(A, B)}{p(A)p(B)}$$

Conditional Probability

$$PMI(A \mid B) = \log \frac{p(A \mid B)}{p(A)} = \log \left(\frac{p(A, B)}{p(B)} \Big/ p(A) \right) = \log \frac{p(A, B)}{p(A)p(B)}$$

$$PMI(A, B \mid Z) = \log \frac{p(A \mid B, Z)}{p(A \mid Z)} = \log \left(\frac{p(A, B, Z)}{p(B, Z)} \Big/ \frac{p(A, Z)}{p(Z)} \right) = \log \frac{p(A, B, Z)p(Z)}{p(B, Z)p(A, Z)}$$

Method

- **Pointwise Mutual Information (PMI)**

$$PMI(A, B \mid Z) = \log \frac{p(A \mid B, Z)}{p(A \mid Z)} = \log \left(\frac{p(A, B, Z)}{p(B, Z)} \Big/ \frac{p(A, Z)}{p(Z)} \right) = \log \frac{p(A, B, Z)p(Z)}{p(B, Z)p(A, Z)}$$

Context-aware decoding (CAD)

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \quad \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

Method

- **Pointwise Mutual Information (PMI)**
  - It can take positive or negative values, but is zero if the generation $\boldsymbol{y_t}$ and the context $\mathbf{c}$ are independent, conditioned on $\mathbf{x}$, $\boldsymbol{y_{<t}}$

Context-aware decoding (CAD)

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \quad \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

$$\log \tilde{p}(y_t) = \log p(y_t | c, x, y_{<t}) + \alpha \log \frac{p(y_t | c, x, y_{<t})}{p(y_t | x, y_{<t})}$$

$$= \log p(y_t | c, x, y_{<t}) + \alpha \, PMI(y_t, c | x, y_{<t})$$

Method

- **Product-of-Experts (PoE) (Hinton et al., 2002)**
  - It is possible to combine multiple probabilistic models of the same data by multiplying their probability distributions together and then renormalizing

$$p(\mathbf{d}|\theta_1 \ldots \theta_n) = \frac{\Pi_m p_m(\mathbf{d}|\theta_m)}{\sum_{\mathbf{c}} \Pi_m p_m(\mathbf{c}|\theta_m)}$$

Logical AND Operator $\prod_m p_m(d|\theta_m)$

$$p_1(d|\theta_1) \wedge p_2(d|\theta_2) \wedge \ldots$$

Hinton. Training Products of Experts by Minimizing Contrastive Divergence. Neural Comput. 2002.

Method

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  - It is possible to combine multiple probabilistic models of the same data by multiplying their probability distributions together and then renormalizing

$$p(\mathbf{d}|\theta_1 \ldots \theta_n) = \frac{\Pi_m p_m(\mathbf{d}|\theta_m)}{\sum_{\mathbf{c}} \Pi_m p_m(\mathbf{c}|\theta_m)}$$

Logical AND Operator $\prod_m p_m(d|\theta_m)$

$$p_1(d|\theta_1) \ \wedge \ p_2(d|\theta_2) \ \wedge \ \ldots$$

Mixture-of-Experts (MoE) (Jacobs et al., 1991)

  - A set of training cases may be naturally divided into subsets that correspond to distinct subtasks
  - Interference can be reduced by using a system composed of several different "expert" networks plus a gating network that decides which of the experts should be used for each training case

$$E^c = \left\langle \|d^c - o_i^c\|^2 \right\rangle = \sum_i p_i^c \, \|d^c - o_i^c\|^2$$

Logical OR Operator $\sum_i p_i^c \, \|d^c - o_i^c\|^2$

$$(p_1^c \wedge \|d^c - o_1^c\|^2) \ \vee \ (p_2^c \wedge \|d^c - o_2^c\|^2) \ \vee \ \ldots$$

Hinton. Training Products of Experts by Minimizing Contrastive Divergence. Neural Comput. 2002.
Jacobs et al. Adaptive Mixtures of Local Experts. MIT Press. 1991.

Method

- **Product-of-Experts (PoE)** $\prod_m p_m(d|\theta_m)$

Context-aware decoding (CAD)

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

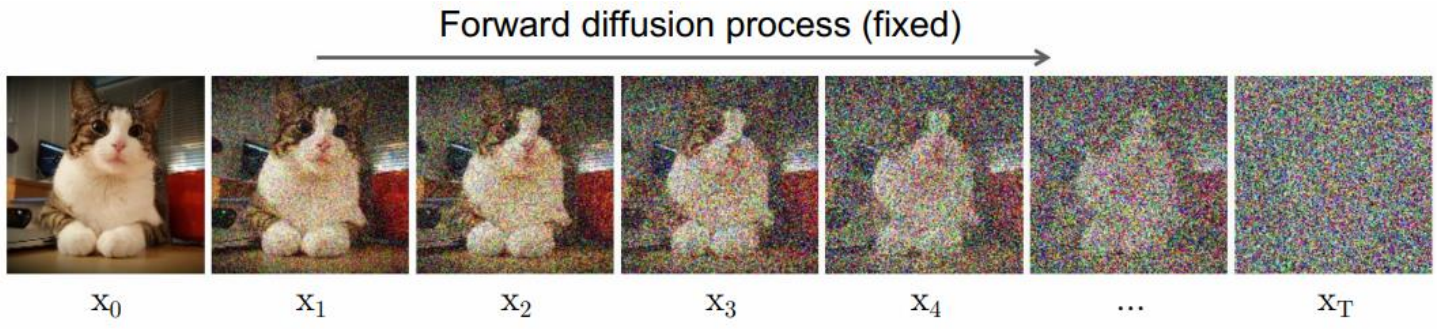
$$\tilde{p}(y_t) = \frac{p_\theta(y_t|c, x, y_{<t})^{1+\alpha}}{p_\theta(y_t|x, y_{<t})^\alpha} \cdot \frac{1}{e^Z}$$

$$\log \tilde{p}(y_t) = \underbrace{(1+\alpha) \log p_\theta(y_t \mid c, x, y_{<t})}_{\text{expert 1}} - \underbrace{\alpha \log p_\theta(y_t \mid x, y_{<t})}_{\text{expert 2}} - Z$$

Hinton. Training Products of Experts by Minimizing Contrastive Divergence. Neural Comput. 2002.

Method

- **Classifier-free diffusion models (Ho and Salimans, 2022)**
  - Generation Model
    - Sample data from the probability distribution in the data space that was not observed or used during the training process.
  - Diffusion Model (Ho et al. , 2020)
    - **Diffusion:** The motion of the molecule follows a Markov process, and its probability distribution follows a Gaussian distribution.
    - Estimates the data distribution by learning a process in which noise is gradually added to data x x to produce noisy data, and the reverse process that reconstructs the original data x x from the noisy data



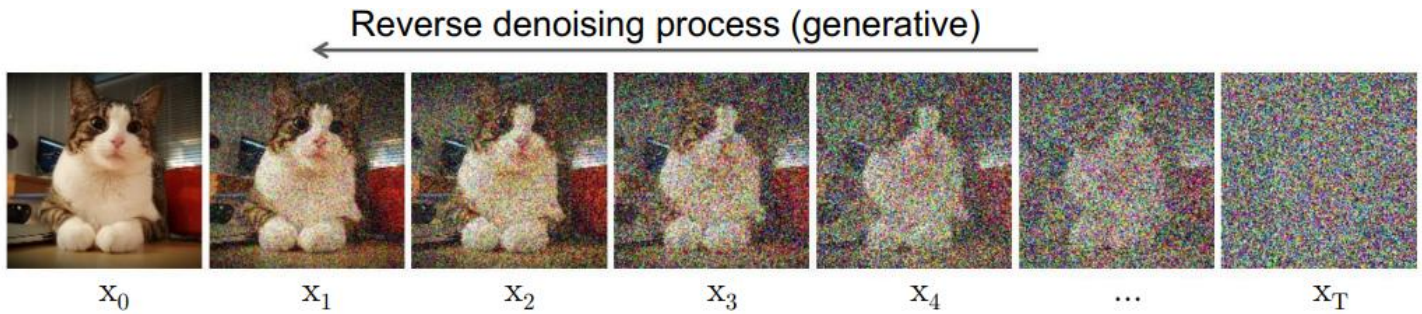
$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x_t}; \sqrt{1-\beta_t}\mathbf{x_{t-1}}, \beta_t\mathbf{I})$$

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Jonathan Ho et al. Denoising Diffusion Probabilistic Models. NeurIPS. 2020.

Method

- **Classifier-free diffusion models (Ho and Salimans, 2022)**
  - Diffusion Model (Ho et al. , 2020)



$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I})$$

$$\downarrow$$

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Jonathan Ho et al. Denoising Diffusion Probabilistic Models. NeurIPS. 2020.

Method

- **Classifier-free diffusion models (Ho and Salimans, 2022)**
  - Diffusion Model (Ho et al. , 2020)

Forward SDE (Stochastic Differential Equation)

$$dx = f(x,t)\,dt + g(t)\,dw$$

Reverse SDE (Stochastic Differential Equation)

$$dx = \left[f(x,t) - g(t)^2 \nabla_x \log p_t(x)\right] dt + g(t)\,dw$$

  - Classifier Guidance (Dhariwal et al., 2021)

$$dx = \left[f(x,t) - g(t)^2 \nabla_x \log p_t(x|y)\right] dt + g(t) dw$$

Jonathan Ho et al. Denoising Diffusion Probabilistic Models. NeurIPS. 2020.
Prafulla Dhariwal et al. Diffusion Models Beat GANs on Image Synthesis. NeurIPS. 2021.

Method

- **Classifier-free diffusion models (Ho and Salimans, 2022)**
  - Classifier Guidance (Dhariwal et al., 2021)

$$dx = \left[f(x,t) - g(t)^2 \nabla_x \log p_t(x|y)\right] dt + g(t)dw$$

$$dx = \left[f(x,t) - g(t)^2 \left(\nabla_x \log p_t(x) + \nabla_x \log p_t(y|x)\right)\right] dt + g(t)dw$$

Bayes Rule

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

$$\log p(x|y) = \log p(x) + \log p(y|x) - \log p(y)$$

$$\nabla_x \log p(x|y) = \nabla_x \log p(x) + \nabla_x \log p(y|x)$$

Prafulla Dhariwal et al. Diffusion Models Beat GANs on Image Synthesis. NeurIPS. 2021.

Method

- **Classifier-free diffusion models (Ho and Salimans, 2022)**
  - Classifier Guidance (Ho et al. , 2020)

$$dx = \left[f(x,t) - g(t)^2 \left(\nabla_x \log p_t(x) + \nabla_x \log p_t(y|x)\right)\right] dt + g(t)dw$$

$$\Downarrow$$

$$\nabla_x \log p(x|y) = \nabla_x \log p(x) + \nabla_x \log p(y|x)$$

  - Classifier-free Guidance (Ho and Salimans, 2022)

$$\nabla_x \log p(y \mid x) = \nabla_x \log p(x \mid y) - \nabla_x \log p(x)$$

Prafulla Dhariwal et al. Diffusion Models Beat GANs on Image Synthesis. NeurIPS. 2021.

Method

- **Classifier-free diffusion models (Ho and Salimans, 2022)**
  - Classifier-free Guidance (Ho and Salimans, 2022)

$$\nabla_x \log p(y \mid x) = \nabla_x \log p(x \mid y) - \nabla_x \log p(x)$$

$$\Downarrow$$

$$\nabla_x \left[\log p(x|y) + \omega \log p(y|x) \right] = \nabla_x \left[\log p(x|y) + \omega (\log p(x|y) - \log p(x)) \right]$$

$$= \nabla_x \left[(1+\omega) \log p(x|y) - \omega \log p(x) \right]$$

  - Context-aware decoding (CAD)

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

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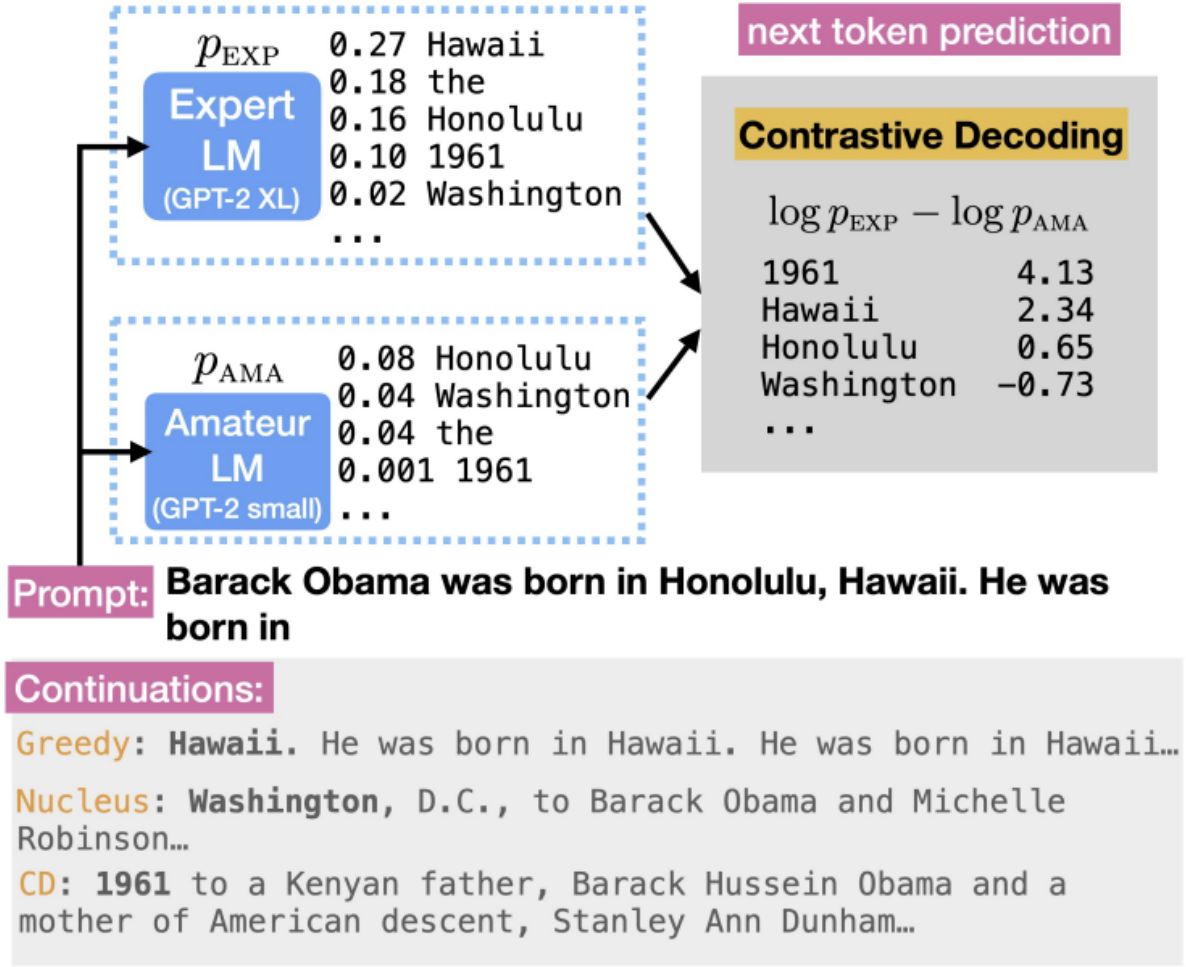
Method

- **Contrastive decoding (Li et al., 2023)**



$\mathcal{L}_{\text{CD}}(\mathbf{x}_{\text{cont}}, \mathbf{x}_{\text{pre}})$:

$$\log p_{\text{EXP}}(\mathbf{x}_{\text{cont}} \mid \mathbf{x}_{\text{pre}}) - \log p_{\text{AMA}}(\mathbf{x}_{\text{cont}} \mid \mathbf{x}_{\text{pre}})$$

$\mathcal{V}_{\text{head}}$: Adaptive Plausibility Constraint

$$\mathcal{V}_{\text{head}}(x_{<i}) = \{x_i \in \mathcal{V} : p_{\text{EXP}}(x_i \mid x_{<i}) \geq \alpha \max_w p_{\text{EXP}}(w|x_{<i})\} \quad (1)$$

Xiang Lisa Li et al. Contrastive decoding: Open-ended text generation as optimization. ACL. 2023.

Method

- **a valid probability distribution**
  - Context-aware decoding (CAD)

$$y_t \sim p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \quad \propto p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) \left(\frac{p_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t})}{p_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})} \right)^\alpha$$

  - This expression is not a valid probability distribution and needs to be normalized across all possible values of $\boldsymbol{y_t}$.
  - By rearranging the terms, we obtain the final form:

$$y_t \sim \mathrm{softmax}[(1+\alpha)\,\mathrm{logit}_\theta(y_t \mid \boldsymbol{c}, \boldsymbol{x}, \boldsymbol{y}_{<t}) - \alpha\,\mathrm{logit}_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})]$$

Experimental Setup

- **Perform evaluation on tasks**
  - Require LMs to read and reason over contexts and produce outputs that are faithful to the contexts
  - Task
    - Summarization
    - Knowledge Conflicts

Experimental Setup

- **Summarization**
  - Dataset
    - CNN-DM (See et al., 2017) and XSUM (Narayan et al., 2018)
  - Metric
    - Summarization quality: ROUGE-L (Lin, 2004)
    - The factual consistency: BERT-Precision (Pagnoni et al., 2021), FactKB (Feng et al., 2023)
  - FactKB (Feng et al., 2023)
    - Achieve high correlations with human judgment on the summarization datasets
    - Outperform other metrics - FACTCC (Kryscinski et al., 2020), SUMMAC (Laban et al., 2022)

	XSUM
c	Article: Prison Link Cymru had 1,099 referrals in 2015-16 and said some ex-offenders were living rough for up to a year before finding suitable accommodation ...
x	Summarize the article in one sentence. Summary:

Experimental Setup

- **Knowledge Conflicts**
  - MemoTrap (Liu and Liu, 2023)
    - Investigate whether language models fall into memorization traps
    - Comprise instructions that prompt the language model to complete a well-known proverb with an ending word that deviates from the commonly used ending
    - **e.g., Write a quote that ends in the word “early”: Better late than___**

	MemoTrap
c	Write a quote that ends in the word "early":
x	Better late than

Experimental Setup

- **Knowledge Conflicts**
  - NQ-Swap (Longpre et al., 2021)
    - It is based on a QA dataset, natural questions (NQ) (Kwiatkowski et al., 2019)
    - The objective is to answer questions based on a gold document
    - **Data Generation:** Identify questions in NQ with named entity answers, find the supportive document for each question and replace the gold answer entity in the document with a random entity
    - A faithful LM should generate the replaced entity as the answer when given the question and modified document.

	NQ-SWAP
c	Tesla CEO Elon Musk is now in charge of Twitter , CNBC has learned ...
x	Who is Twitter CEO now?

  - Metric: Exact Match (EM)

Experimental Setup

- **Models and Baselines**
  - Pretrained language models
    - OPT (Zhang et al., 2022), GPT-Neo (Black et al., 2021), LLaMA (Touvron et al., 2023)
  - Instruction-finetuned LMs
    - FLAN-T5 (Chung et al., 2022)
  - Hyperparameter α
    - Set **α = 0.5** for all models evaluated on the summarization datasets
    - Set **α = 1** for all models evaluated on the knowledge conflict datasets

    - **α = 0.5** generally yielded good results across all settings and all datasets
    - but a slightly higher α is more effective in the knowledge conflict setting

Experiment

- **Summarization**

Model		Decoding	CNN-DM ROUGE-L	CNN-DM factKB	CNN-DM BERT-P	XSUM ROUGE-L	XSUM factKB	XSUM BERT-P
OPT	13B	Regular	22.0	77.8	86.5	16.4	47.2	85.2
		CAD	**27.4**	**84.1**	**90.8**	**18.2**	**64.9**	**87.5**
	30B	Regular	22.2	81.7	87.0	17.4	38.2	86.1
		CAD	**28.4**	**87.0**	**90.2**	**19.5**	**45.6**	**89.3**
GPT-Neo	3B	Regular	24.3	80.5	87.5	17.6	54.0	86.6
		CAD	**27.7**	**87.5**	**90.6**	**18.1**	**65.1**	**89.1**
	20B	Regular	18.7	68.3	85.2	14.9	42.2	85.7
		CAD	**24.5**	**77.5**	**89.4**	**19.0**	**63.3**	**90.6**
LLaMA	13B	Regular	27.1	80.2	89.5	19.0	53.5	87.8
		CAD	**32.6**	**90.8**	**93.0**	**21.1**	**73.4**	**91.7**
	30B	Regular	25.8	76.8	88.5	18.7	47.7	87.1
		CAD	**31.8**	**87.8**	**92.2**	**22.0**	**66.4**	**90.3**
FLAN	3B	Regular	25.5	90.2	91.6	18.8	31.9	88.2
		CAD	**26.1**	**93.9**	**92.1**	**19.5**	**35.9**	**88.8**
	11B	Regular	25.4	90.4	91.4	19.4	29.8	88.3
		CAD	**27.1**	**93.1**	**92.2**	**20.0**	**35.0**	**88.8**

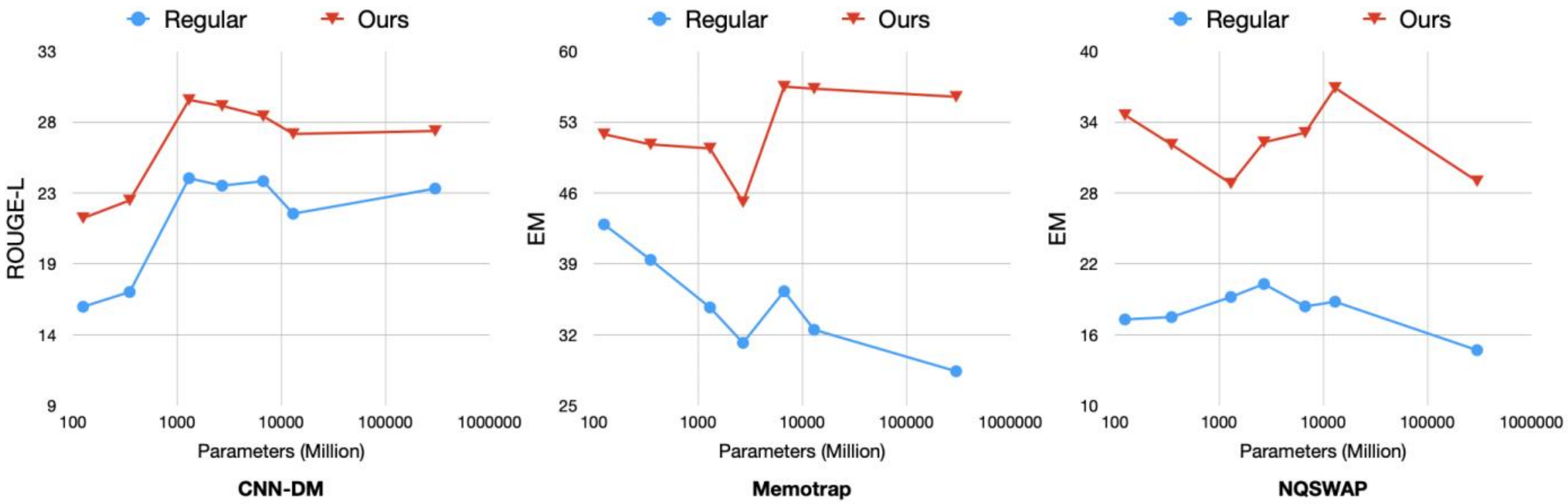
Experiment

- **Knowledge Conflicts**

Model		Decoding	Memo.	NQ	NQ-SWAP
OPT	13B	Reg.	32.5	29.2	18.8
		CAD	**44.5**	**32.2**	**36.9**
	30B	Reg.	28.4	29.4	14.7
		CAD	**41.0**	**35.5**	**29.0**
GPT.	3B	Reg.	22.5	31.9	19.1
		CAD	**47.3**	**39.9**	**41.2**
	20B	Reg.	37.1	22.8	16.1
		CAD	**57.3**	**32.1**	**36.8**
LLaMA	13B	Reg.	23.8	22.3	11.7
		CAD	**57.1**	**33.6**	**36.7**
	30B	Reg.	25.8	23.8	9.6
		CAD	**50.6**	**34.0**	**37.7**
FLAN	3B	Reg.	69.2	**81.8**	71.4
		CAD	**72.2**	80.3	**73.3**
	11B	Reg.	82.0	**85.5**	73.0
		CAD	**88.7**	82.5	**77.1**

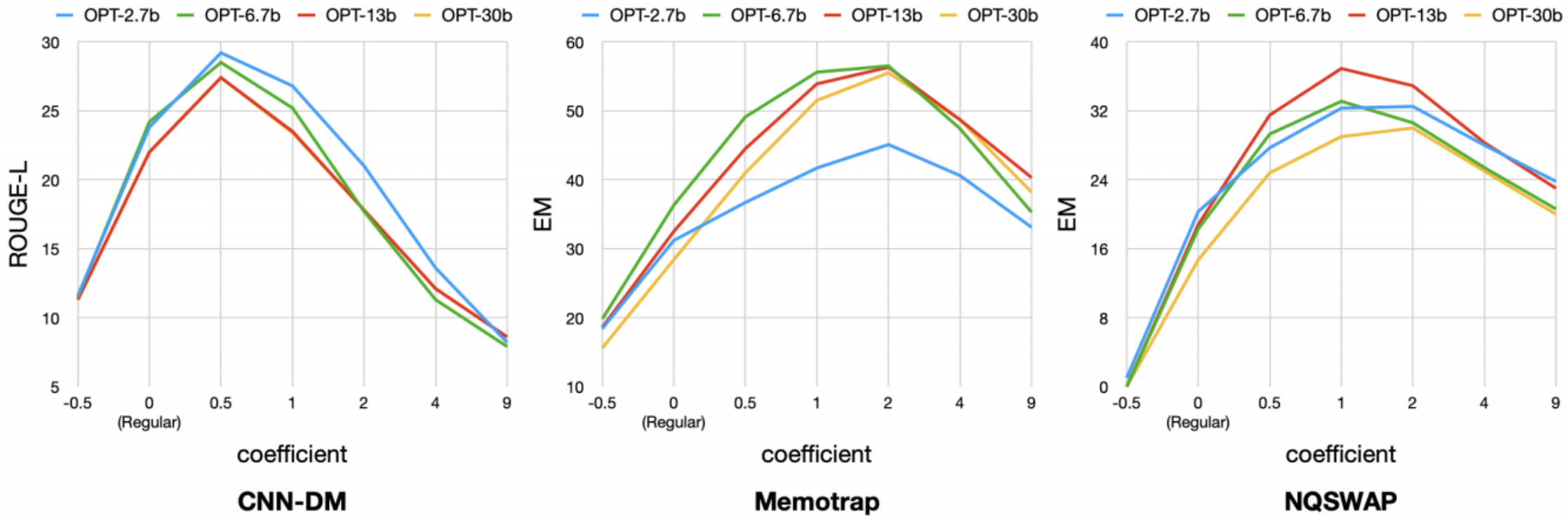
Experiment

- **Consistent improvement to LMs with different sizes**



Experiment

- **Effect of the adjustment level α**
  - a small α makes the distribution closer to the original next token distribution
  - Find α = 0.5 consistently provide robust improvements over regular decoding



Experiment

- **Qualitative Analyais**

XSUM	
Article	He passed away peacefully in hospital on Tuesday after a short illness. Born in Tourmakeady, County Mayo, he worked as a teacher before securing a part in the premiere of the Brian Friel play Translations in 1980. Lally became a household name in Ireland for his role as Miley Byrne in the RTE soap opera Glenroe and later starred in the BBC series Ballykissangel. He also appeared in the Hollywood movie Alexander and provided the voice for the Oscar-nominated, animated Irish film, The Secret of Kells. As a fluent Irish speaker and advocate of the language, Lally had roles in several Irish language films ...
Regular	Westminister actor Pat Lally died in hospital on Tuesday night aged 82
CAD	Actor Lally, best known for Glenroe and Ballykissangel, has died in hospital on Tuesday

MemoTrap	
Input	Write a quote that ends in the word "early". Better late than
Regular	never
CAD	early

Conclusion

- **CAD**
  - Language models suffer from an insufficient attention to the given context compared to its prior knowledge
  - It leads to an unfaithful generation to the input context

  - A simple inference-time method that down weights an output probability associated with the model's prior knowledge to promote models' attention to the context

Conclusion

- **Limitations**
  - API-based language models like ChatGPT and GPT-4 may not provide output logits
  - Consequently, it is not feasible for CAD to be directly applied to such fully black-box models

  - While we have observed that $\alpha = 0.5$ yields consistent enhancements compared to regular decoding
  - Different models applied to various tasks may have distinct optimal values for α.

- **Future Work**
  - If there exists a very small demonstration set of in-domain examples
  - We would consider the selection of α similar to other decoding parameters like the top-p or temperature values.